

Présentation rapport TER

Approche moderne du rendu différentiel

Julien LEFEBVRE et Corentin VAILLANT

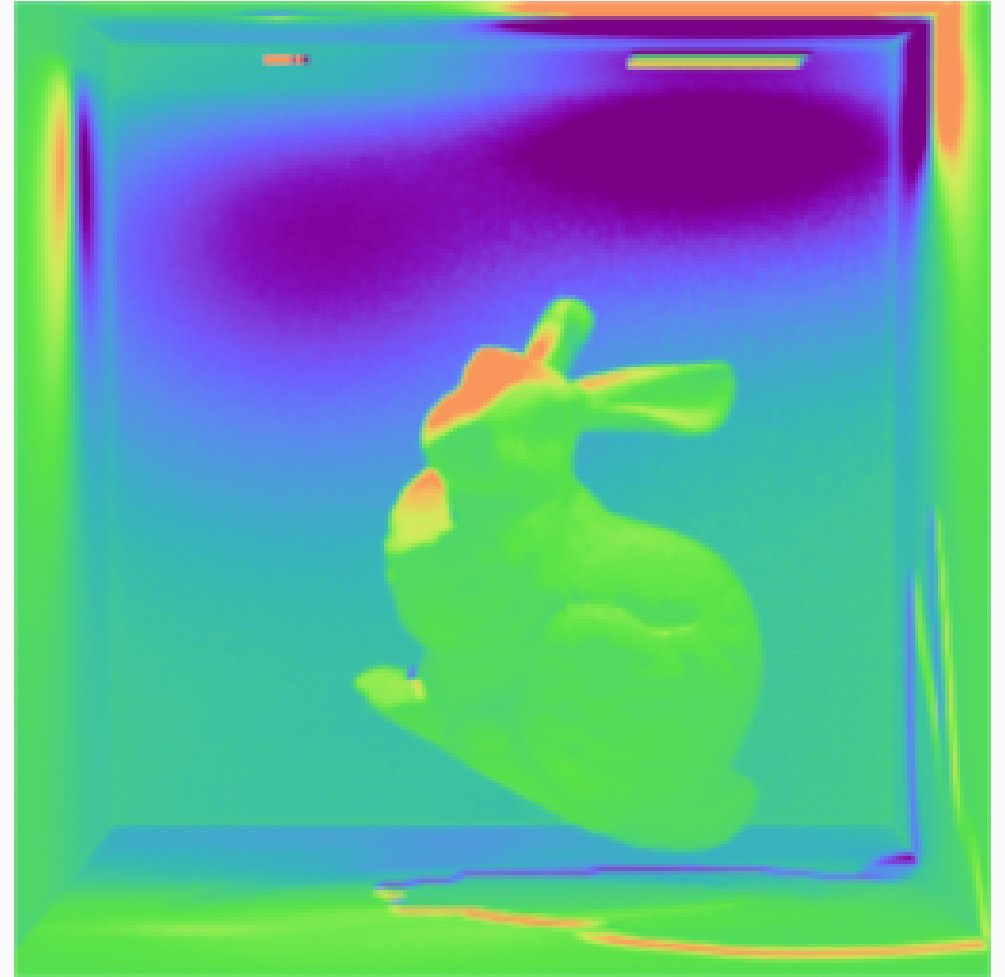
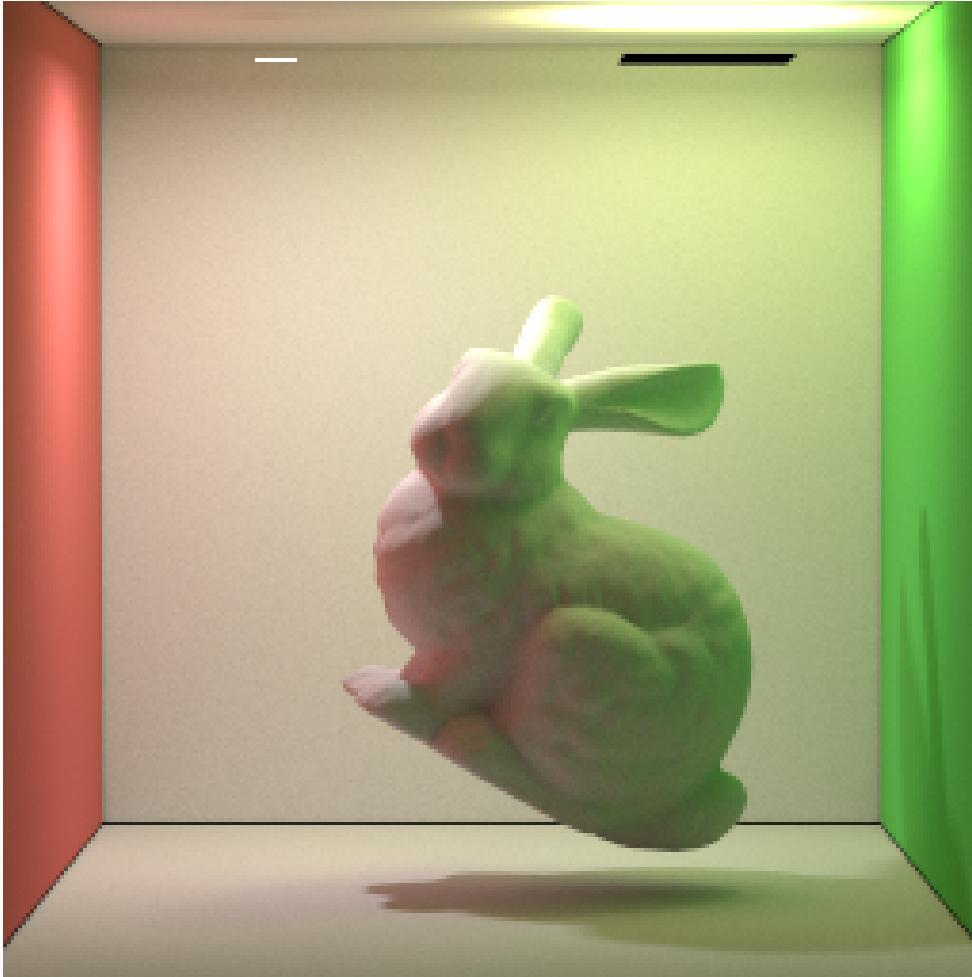
2026

MIDL - FSI Université de Toulouse

1. Introduction



1. Introduction



2. Le Rendu

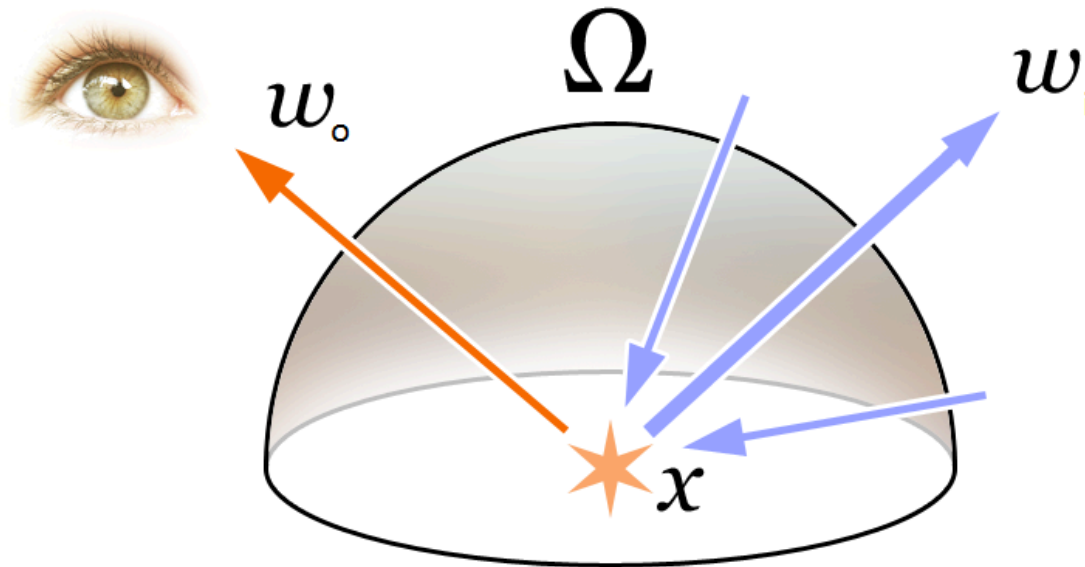


2.1 L'équation de Kajiya

Définition 2.1. (Equation du Rendu)

$$L_i(x, \omega_i) = L_o(vis(x, \omega_i), -\omega_i) \quad (1)$$

$$L_o(x, \omega_o) = L_e(x, \omega_o) + \int_{\Omega} L_i(x, \omega_i) f_i(x, \omega_o, \omega_i) \overline{\cos(\theta_i)} d\omega_i \quad (2)$$

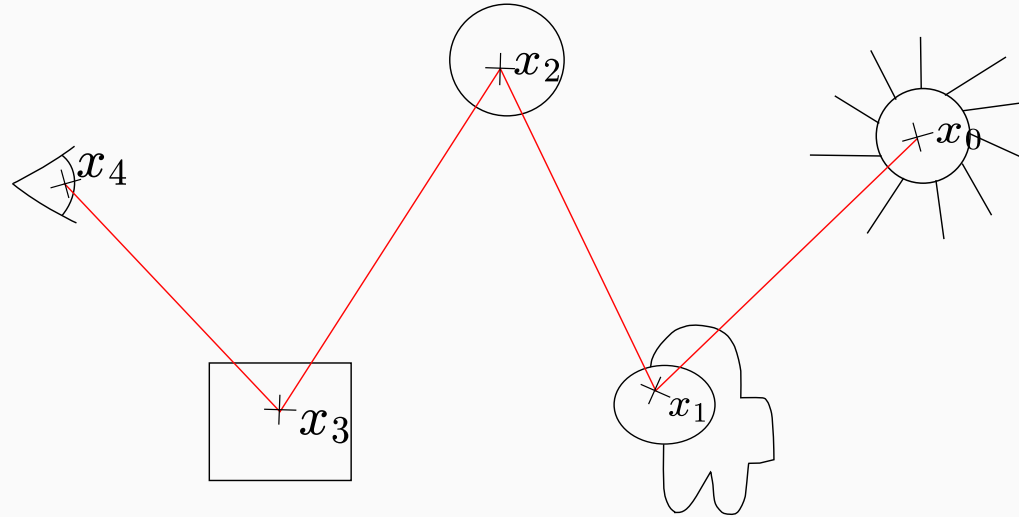


2.1 L'équation de Kajiya

```
Lumière L_i(Rayon r, int depth){
    Intersection inter = TracerRayon(r);
    if(!inter)
        return L_e(r); // Le rayon n'intersecte pas
    Vec3 wo = -r.direction;
    Lumière L = inter.L_e(wo);
    if(depth >= MAX_DEPTH)
        return L;
    Vec3 wi = DirectionAléatoireUniforme();
    Lumière f_cos = inter.bsdf(wo,wi) * abs(dot(wi, inter.normal));
    if(f_cos == 0)
        return L;
    r = {inter.point, wi};
    return L + f_cos * L_i(r, depth +1) / (1/(4*pi));
}
```

2.2 L'équation de Veach

$$\bar{x} := (x_0, x_1, \dots, x_n) \in \mathcal{M}^{n+1}$$



$$d\mu(\bar{x}) := \prod_{n=0}^N dA(x_n)$$

2.2 L'équation de Veach

Définition 2.2. (Équation du rendu sur l'espace des chemins)

$$I = \int_{\Omega} f_{\pi}(\bar{x}) d\mu(\bar{x}) \quad (3)$$

$$f_{\pi}(\bar{x}) := \left(\prod_{n=0}^{N-1} g(x_{n+1} : x_{n-1}, x_n) \right) W_e(x_n \rightarrow x_{N-1}) \quad (4)$$

$$g(x_{n+1} : x_{n-1}, x_n) := f_s(x_{n-1} \rightarrow x_n \rightarrow x_{n+1}) G(x_n \leftrightarrow x_{n+1}) \quad (5)$$

$$G(x_n \leftrightarrow x_{n+1}) := \mathbb{V}(x_n \leftrightarrow x_{n+1}) G_0(x_n \leftrightarrow x_{n+1}) \quad (6)$$

$$G_0(x_n \leftrightarrow x_{n+1}) := \frac{|\vec{n}_{x_n} \cdot \omega_n| |\vec{n}_{x_{n+1}} \cdot (-\omega_n)|}{\|x_{n+1} - x_n\|^2} \quad (7)$$

2.2 L'équation de Veach

```
Lumière L_i(Rayon r){
    Intersection inter = TracerRayon(r);
    if(!inter)
        return L_e(r); // Le rayon n'intersecte pas
    Lumière L = inter.L_e(-r.direction);
    Lumière T = 1; // Débit
    while(T > EPSILON){
        Point p = PointLumièreAléatoire();
        Vec3 wi = normaliser(p - inter.point);
        Lumière alpha_direct =

        ...
    }
}
```

3. La différentiation



3.1 Différentiation dans l'espace des chemins

$$\frac{\partial I}{\partial \pi} = \frac{\partial}{\partial \pi} \int_{\Omega} f_{\pi}(\bar{\mathbf{x}}) d\mu(\bar{\mathbf{x}}) \quad (8)$$

3.1 Différentiation dans l'espace des chemins

$$\frac{\partial I}{\partial \pi} = \frac{\partial}{\partial \pi} \int_{\Omega} f_{\pi}(\bar{\mathbf{x}}) d\mu(\bar{\mathbf{x}}) \quad (8)$$

$$\frac{\partial I}{\partial \pi} \stackrel{?}{=} \int_{\Omega} \frac{\partial f_{\pi}(\bar{\mathbf{x}})}{\partial \pi} d\mu(\bar{\mathbf{x}}) \quad (9)$$

3.1 Différentiation dans l'espace des chemins

$$\frac{\partial I}{\partial \pi} = \frac{\partial}{\partial \pi} \int_{\Omega} f_{\pi}(\bar{\mathbf{x}}) d\mu(\bar{\mathbf{x}}) \quad (8)$$

$$\frac{\partial I}{\partial \pi} \stackrel{?}{=} \int_{\Omega} \frac{\partial f_{\pi}(\bar{\mathbf{x}})}{\partial \pi} d\mu(\bar{\mathbf{x}}) \quad (9)$$

NON !

3.1 Différentiation dans l'espace des chemins

3.1.1 La relation de Reynold

$$\frac{d}{d\pi} \int_{\mathcal{M}} \varphi_{\pi} dA = \underbrace{\int_{\mathcal{M}} \dot{\varphi}_{\pi} dA}_{\text{Intérieur}} + \underbrace{\int_{\Delta\mathcal{M}} \Delta\varphi_{\pi} V_{\Delta\mathcal{M}} d\ell}_{\text{Bordure}} \quad (10)$$

$$\Delta\varphi(x, \pi) := \begin{cases} \varphi(x, \pi) & \text{si } x \in \partial\mathcal{M}(\pi) \\ \varphi^{-}(x, \pi) - \varphi^{+}(x, \pi) & \text{si } x \in \Delta\mathcal{M}(\pi) \end{cases} \quad (11)$$

3.1 Différentiation dans l'espace des chemins

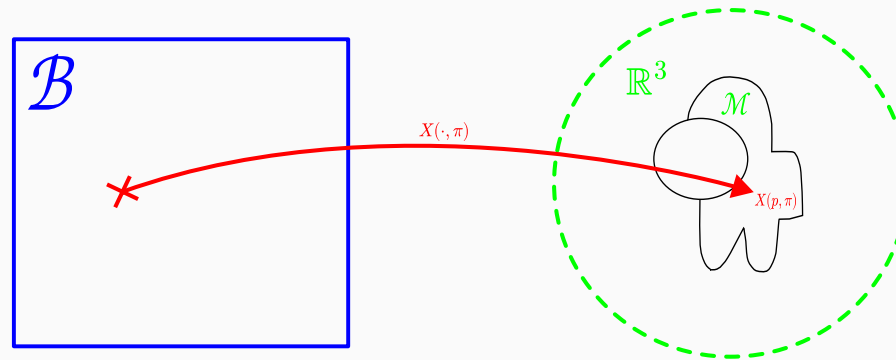
$$\frac{\partial I}{\partial \pi} = \frac{\partial}{\partial \pi} \int_{\Omega} f_{\pi}(\bar{x}) d\mu(\bar{x}) \quad (12)$$

$$\begin{aligned} \frac{\partial I}{\partial \pi} = & \int_{\Omega} \left[\dot{f}(\bar{x}) - f(\bar{x}) \sum_{K=0}^N \kappa(x_K) V(x_K) \right] d\mu(\bar{x}) \\ & + \int_{\partial\Omega} \Delta f_K(\bar{x}) V_{\partial\mathcal{M}_K}(x_K) d\mu'(\bar{x}) \end{aligned} \quad (13)$$

3.1 Différentiation dans l'espace des chemins

$$I = \int_{\hat{\Omega}} \hat{f}(\bar{p}) d\mu(\bar{p}) : \hat{\Omega} := \cup_{N=1}^{\infty} \mathcal{B}^{N+1} \quad (14)$$

$$\hat{f}(\bar{p}) = \left(\prod_{n=0}^{N-1} \hat{g}(p_{n+1}; p_{n-1}, p_n) \right) \widehat{W}_e(p_N \rightarrow p_{N-1}) \quad (15)$$



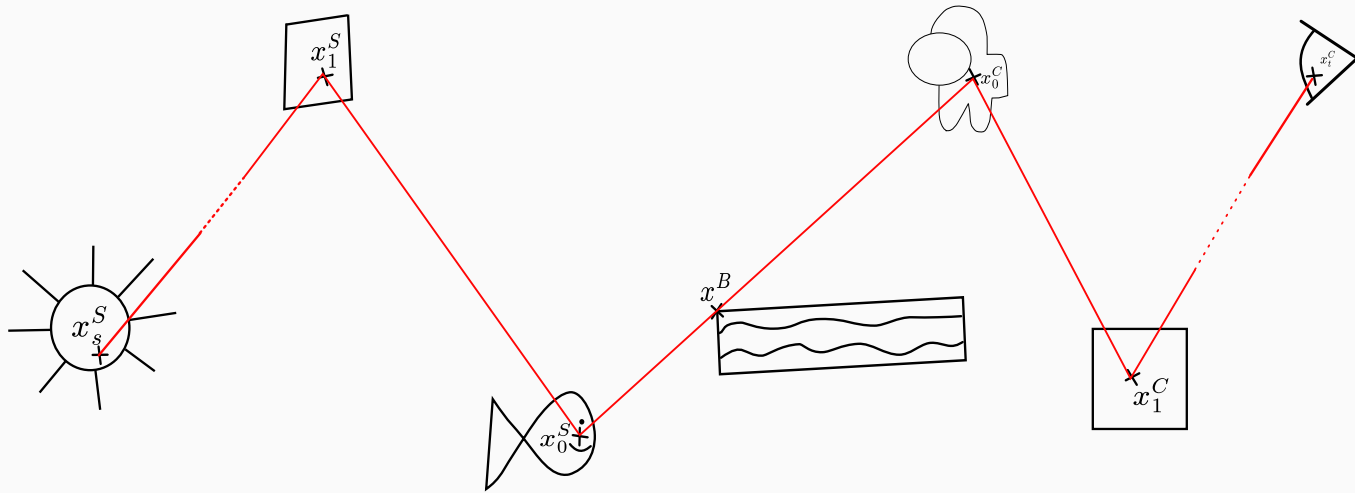
3.1 Différentiation dans l'espace des chemins

$$\frac{\partial I}{\partial \pi} = \frac{\partial}{\partial \pi} \int_{\Omega} f_{\pi}(\bar{\mathbf{x}}) d\mu(\bar{\mathbf{x}}) \quad (16)$$

$$\frac{\partial I}{\partial \pi} = \underbrace{\int_{\hat{\Omega}} (\hat{f})^{\cdot}(\bar{p}) d\mu(\bar{p})}_{\text{Intérieur}} + \underbrace{\int_{\partial \hat{\Omega}} \Delta \hat{f}_K(\bar{p}) V_{\Delta \mathcal{B}_K}(p_K) d\mu'(\bar{p})}_{\text{Bordure}} \quad (17)$$

3.1 Différentiation dans l'espace des chemins

$$\bar{p} := (p_s^S, \dots, p_0^S, p_0^C, \dots, p_t^C) \in \mathcal{B}^{n+1}$$

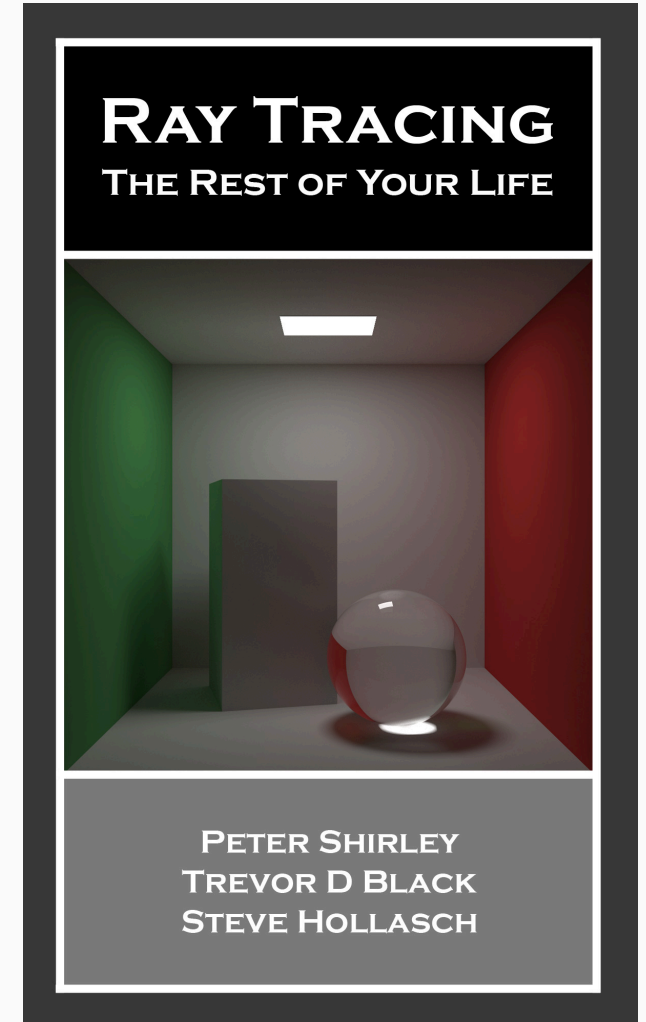
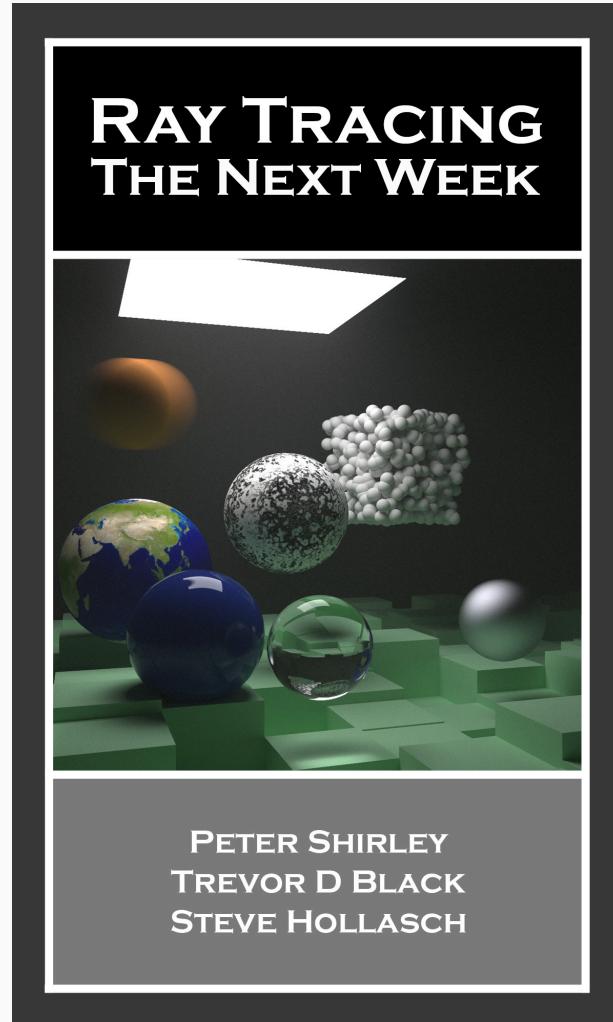
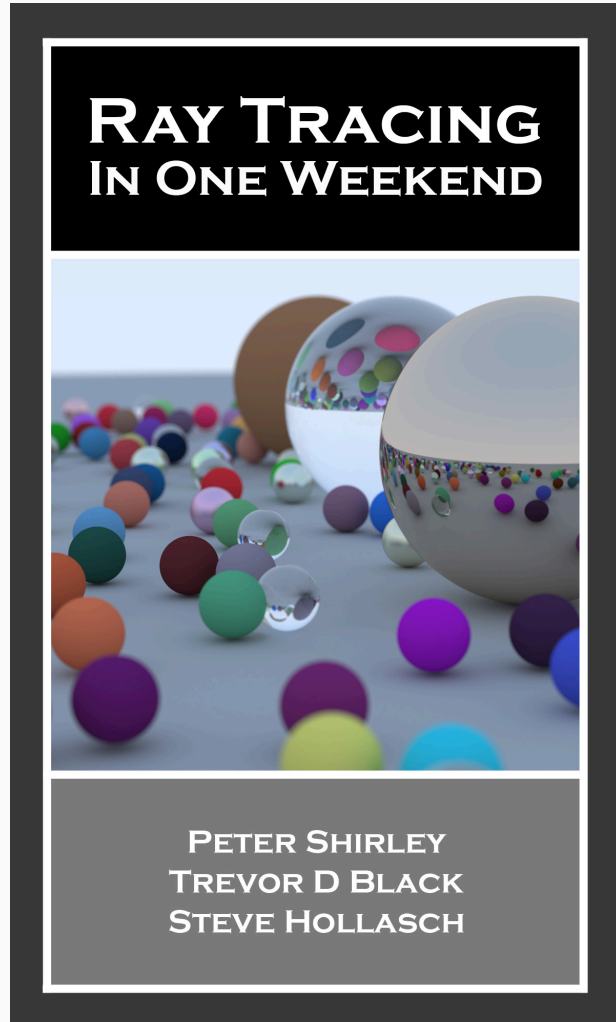


3.1 Différentiation dans l'espace des chemins

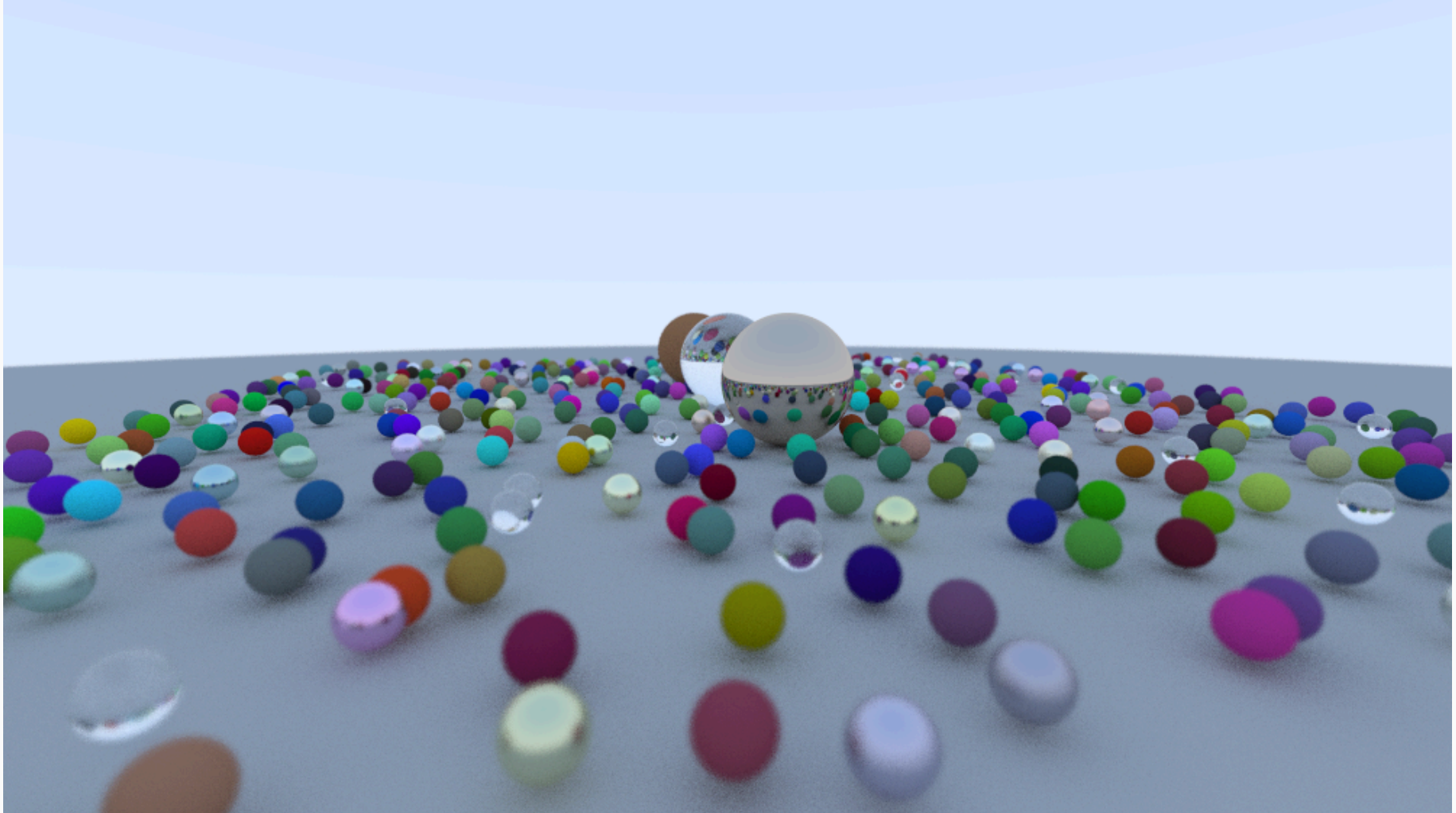
```
LumièreDiff PT_diff_bordure(DistribRayon P, bool direct){
    Rayon [xB,wB] = TirerRayon(P);
    Intersection inter_src = Intersection((xB, -wB));
    Intersection inter_cam = Intersection((xB,wB));
    if(inter_src && inter_cam){
        LumièreDiff TB = fHatB * JB(xB, wB) / P(xB,wB);
        float p0S = VersMatériaux(inter_src.point,pi);
        float p0C = VersMatériaux(inter_cam.point,pi);
        LumièreDiff TS, TC;
        if(direct) TS = inter_src.L_e(Rayon(inter_src.point, inter_cam.point));
        else TS = EstimationCheminSource(p0C,p0S);
        TC = EstimationCheminCamera(p0C,p0S);
        return TS * TB * TC
    }
    else return 0;
}
```

4. Implantation

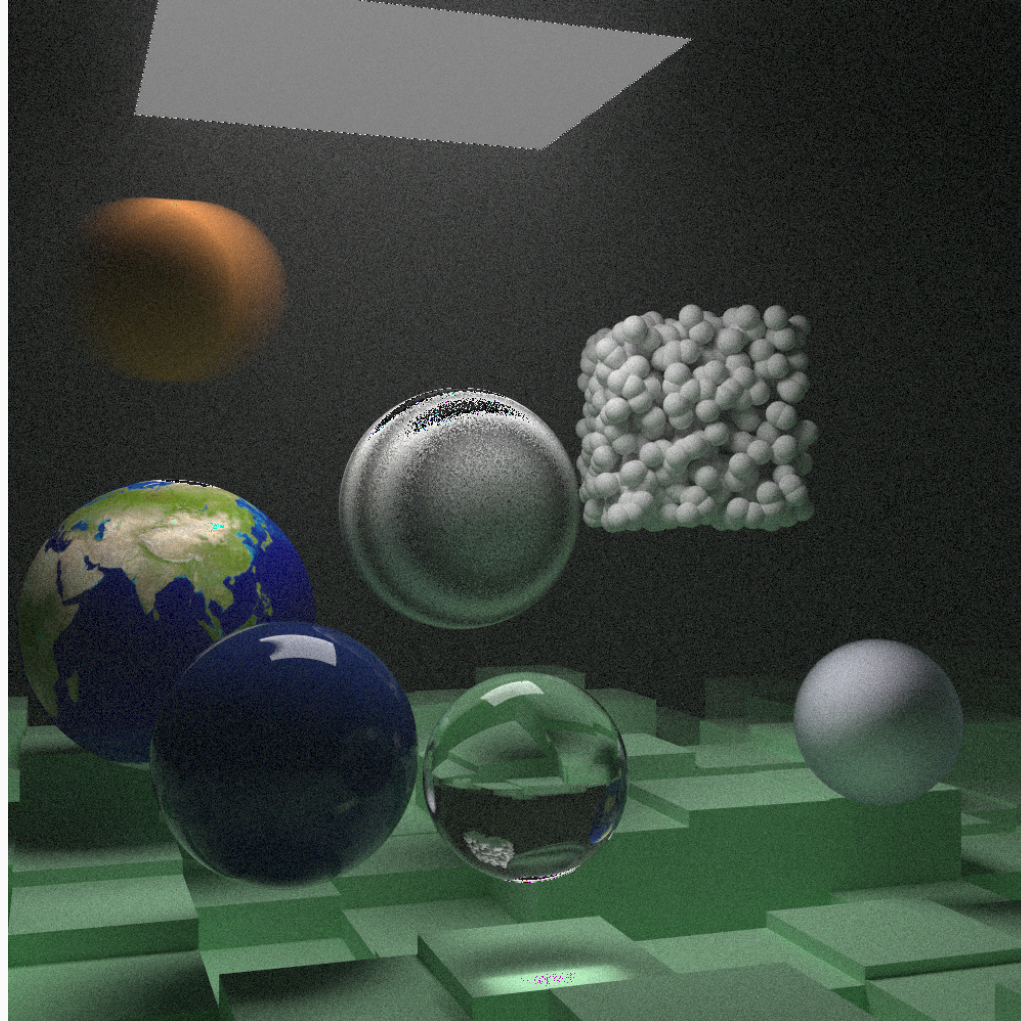
4.1 Ray Tracing in One Weekend



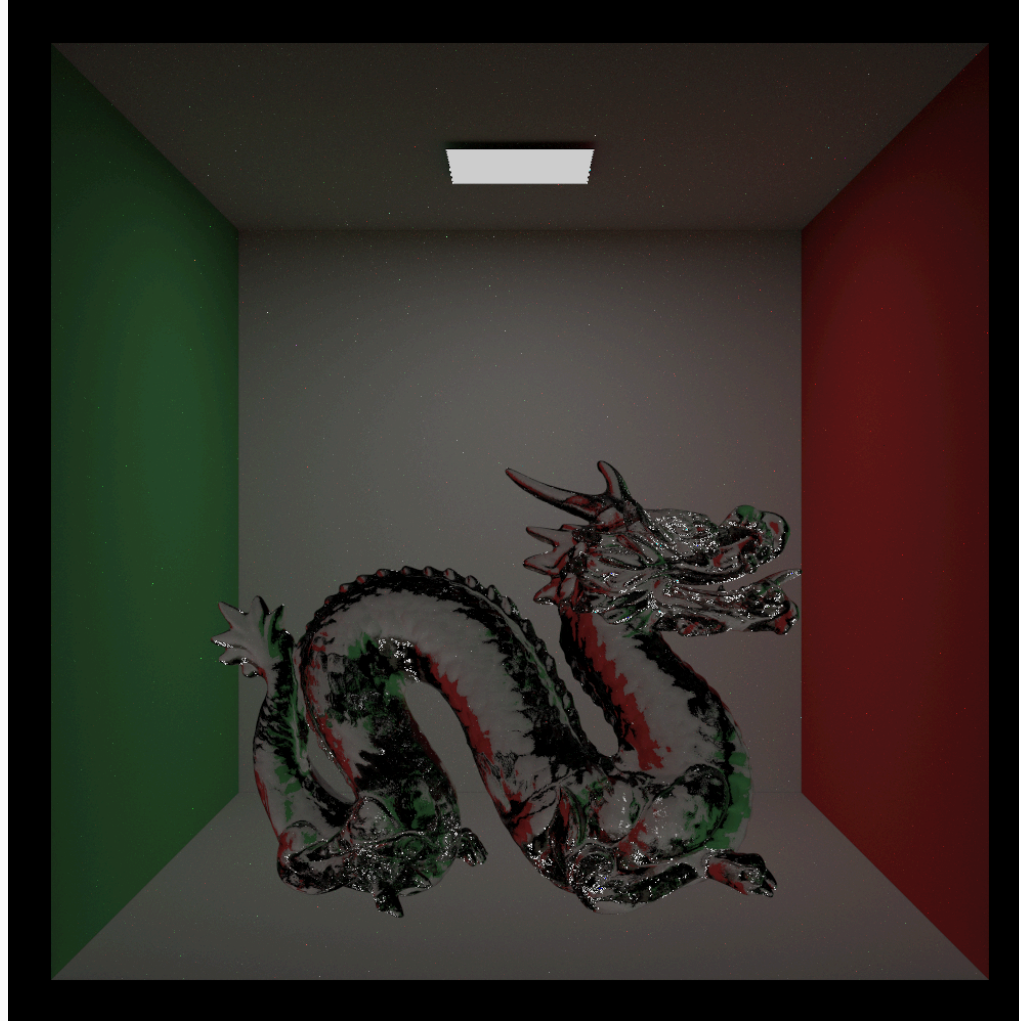
4.1 Ray Tracing in One Weekend



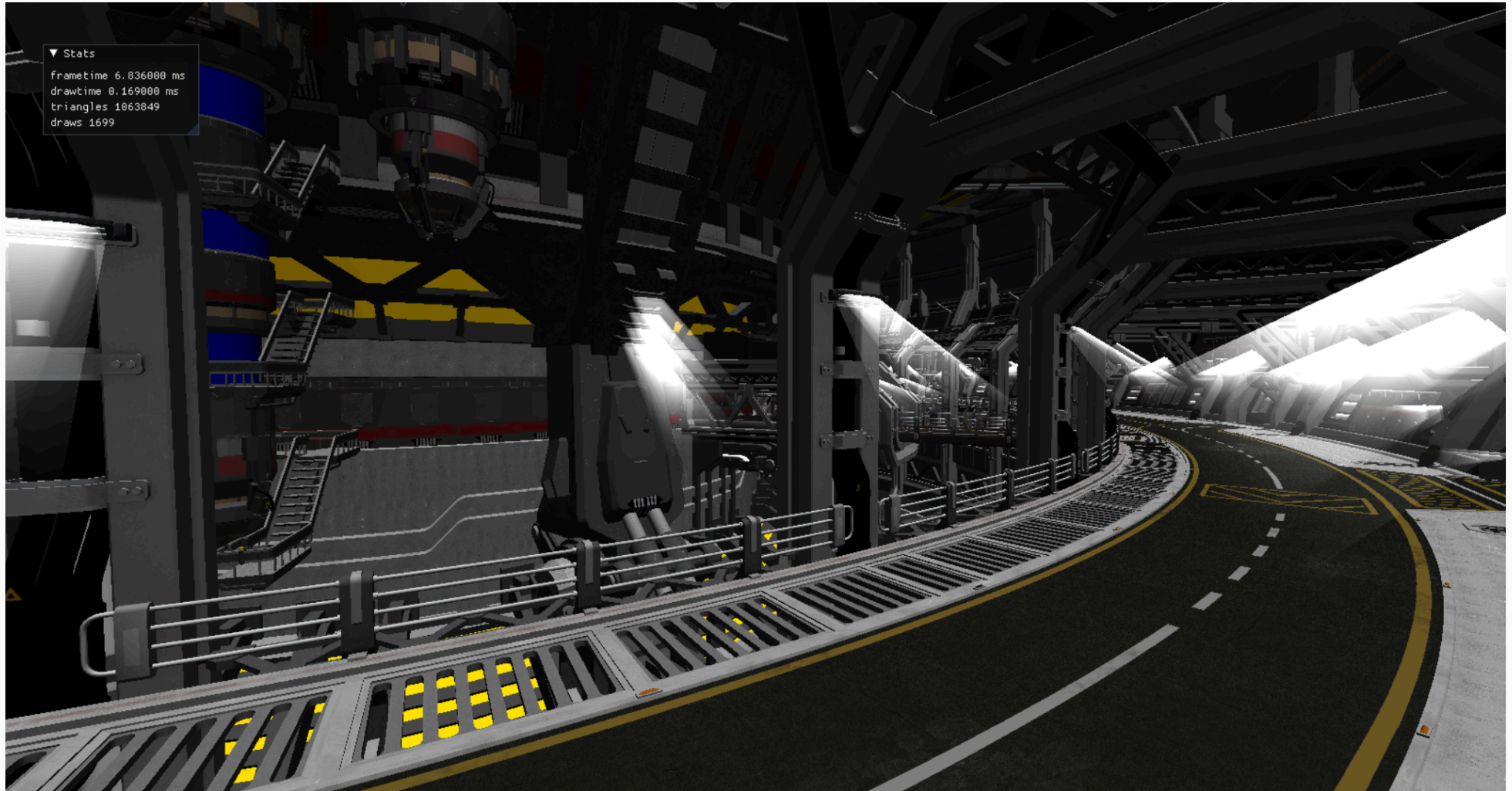
4.1 Ray Tracing in One Weekend



4.1 Ray Tracing in One Weekend



4.2 Vulkan



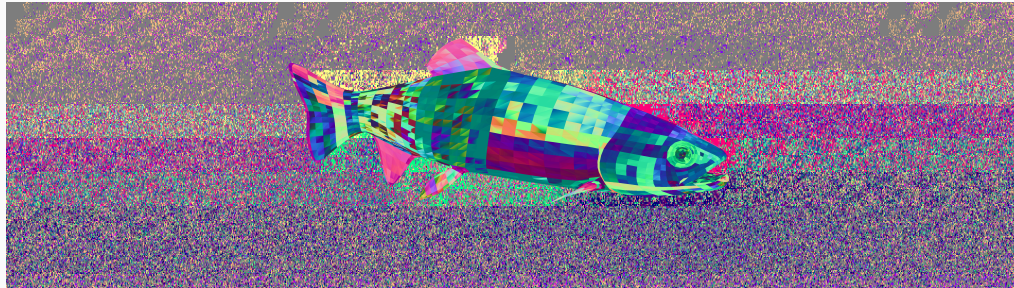
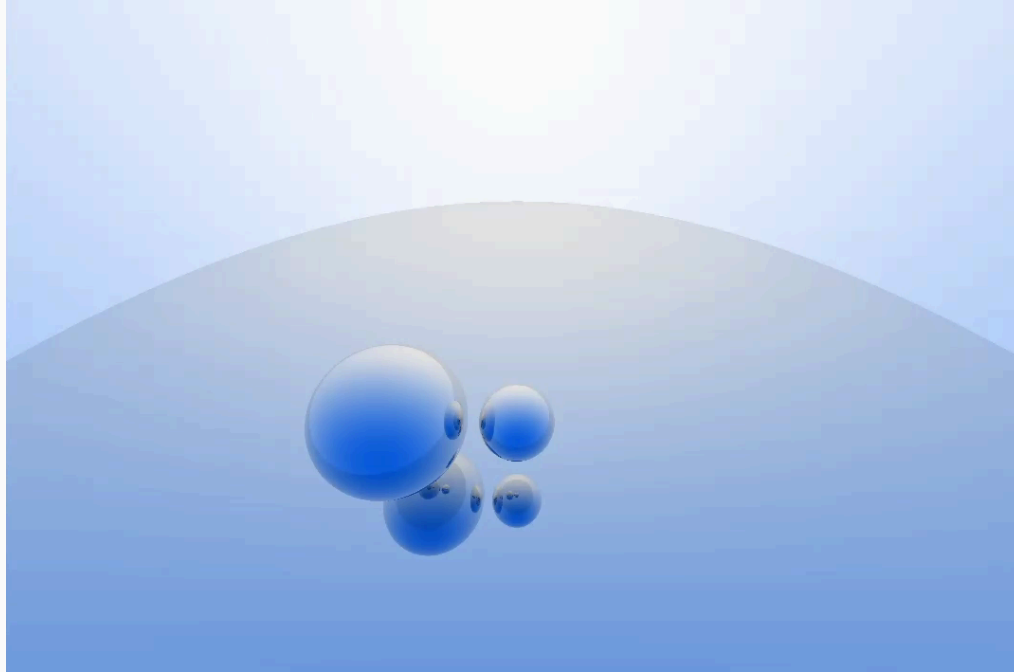


K H R O N O S[®]
G R O U P



Doom Eternal 2020, fait avec Vulkan.

4.2 Vulkan



5. Conclusion

5. Conclusion

TODO